

TOKENIZATION & VOCABULARY CHEATSHEET

AI Engineer Ready · Week 2 — Tokens, Tokenization, and Vocabulary

Golden Rule → text → tokenizer → token IDs → model → cost & context measured in tokens, not words

1. Token vs Tokenization vs Vocabulary

Term	Definition
Token	One unit the model reads — a word, sub-word, punctuation mark, or symbol
Tokenization	The process that splits raw text into tokens
Vocabulary	The fixed list of tokens a specific model knows (each maps to an integer ID)

```
$ text = "unbelievable!"
$ BERT tokenizer -> ['un', '##bel', '##ie', '##vable', '!']
$ each piece maps to an ID from BERT's vocabulary, e.g. [4895, 8331, 2666, 12879, 999]
```

The **##** prefix marks a sub-word continuation ("this piece attaches to the previous token, no space before it") — this is how BERT-style WordPiece tokenizers handle rare or unseen words.

2. Why Tokenization Matters

Reason	Practical impact
Cost model	APIs bill per 1,000 tokens, not per word — a longer prompt = more \$
Context limits	Every model has a max token count for prompt + response combined
Model efficiency	Sub-word tokenization handles rare/unseen words far better than whole-word vocabularies

3. Tokenization Differs Across Models

The same sentence can split differently depending on the model's tokenizer — **BERT**, **GPT-4o-mini**, and **GPT-5** do not share a vocabulary. GPT-4o-mini and GPT-5 use OpenAI's **o200k_base** encoding; tokenizer behavior can evolve across model versions, so always re-check token counts when you switch models.

4. Token Budget Example

```
$ Prompt: 500 tokens
$ Response: 300 tokens
$ Total: 800 tokens ~ $0.001 - $0.003 (depends on model pricing)
```

Pitfall — "1 word = 1 token"

Not true. Short common words are often 1 token, but longer or rare words can split into 2-4+ tokens, and emoji/non-English text can cost even more. Always measure with the model's real tokenizer instead of estimating by word count.

5. Counting Tokens in Code

```
$ import tiktoken
$ enc = tiktoken.encoding_for_model("gpt-4o-mini")
$ tokens = enc.encode("Hello, world!")
$ print(len(tokens)) # -> token count
```